# ANDROID PROGRAMMING PRACTICAL 15

## Module 2 · Runtime Permissions and AndroidManifest security

### Aim

Implement runtime permissions and security handling in AndroidManifest.xml.

**Project name**: PermissionDemo

### Requirements

Android Studio, Java, Android SDK, and an Android Virtual Device or compatible Android phone. Use the package name generated by your project and adjust imports/dependencies to the installed Android Studio template.

### Procedure

**1.** Identify the protected feature (for example, camera) and add the relevant uses-permission declaration above the application tag.

**2.** If the feature is optional, declare uses-feature with android:required="false" as appropriate.

**3.** Add a button that checks ContextCompat.checkSelfPermission().

**4.** If permission is not granted, request it using the Activity Result API or the source-compatible permission request flow.

**5.** Handle granted, denied, and rationale/permanent-denial cases with clear user feedback.

**6.** Run on a device/emulator that supports the permission and test both grant and deny choices.

**7.** Review the manifest for least-privilege permissions and ensure components have appropriate exported settings.

### Implementation checklist

• Keep resource names lowercase and use unique view IDs.

• Make sure every referenced layout, menu, drawable, string, or Java class exists in the project.

• Build the project and resolve compile errors before running it.

• Use Logcat to diagnose runtime errors and permissions or manifest issues.

### Expected result

The app checks/request permissions and handles grant or denial safely.

### Testing / observations

Record the device/API level, the actions tested, the observed output, and any errors corrected. Attach a screenshot of the successful run to your practical record if required by your institution.

### Conclusion

The stated Android concept was implemented and verified through the steps and tests above.